

§7.3.2 典型非线性特性的描述函数

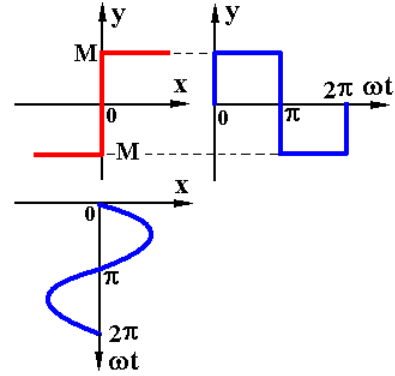
将周期函数 $y(t)$ 展开成傅氏级数形式

$$\begin{aligned} y(t) &= \sum_{n=-\infty}^{\infty} c_n \cdot e^{jn\omega t} = c_0 + \sum_{n=1}^{\infty} [c_n \cdot e^{jn\omega t} + c_{-n} \cdot e^{-jn\omega t}] \\ &= A_0 + \sum_{n=1}^{\infty} [A_n \cos n\omega t + B_n \sin n\omega t] \\ &= A_0 + \sum_{n=1}^{\infty} Y_n \sin(n\omega t + \varphi_n) \end{aligned}$$

其中

$$\begin{cases} A_0 = \frac{1}{2\pi} \int_0^{2\pi} y(t) \cdot d\omega t \\ A_n = \frac{1}{\pi} \int_0^{2\pi} y(t) \cdot \cos n\omega t \cdot d\omega t \\ B_n = \frac{1}{\pi} \int_0^{2\pi} y(t) \cdot \sin n\omega t \cdot d\omega t \end{cases}$$

$$\begin{cases} Y_n = \sqrt{A_n^2 + B_n^2} \\ \varphi_n = \arctan \frac{A_n}{B_n} \end{cases}$$



对于 $y(t)$ 中的基波分量

$$y_1(t) = A_1 \cos \omega t + B_1 \sin \omega t = Y_1 \sin(\omega t + \varphi_1)$$

$$\begin{cases} A_1 = \frac{1}{\pi} \int_0^{2\pi} y(t) \cos \omega t \cdot d\omega t \\ B_1 = \frac{1}{\pi} \int_0^{2\pi} y(t) \sin \omega t \cdot d\omega t \end{cases}$$

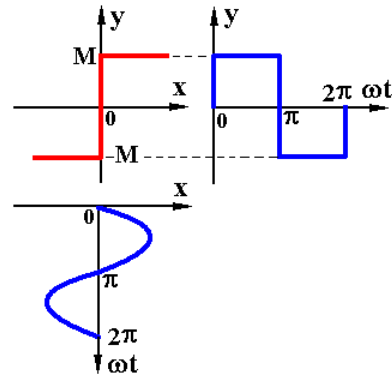
$$\begin{cases} Y_1 = \sqrt{A_1^2 + B_1^2} \\ \varphi_1 = \arctan \frac{A_1}{B_1} \end{cases}$$

1. 理想继电特性的描述函数

$$A_0 = \frac{1}{2\pi} \int_0^{2\pi} y(t) \cdot d\omega t = 0$$

由 $y(t)$ 的奇对称性质 $y(-t) = -y(t)$,

可以确定 $A_i = 0 \quad i = 1, 2, \dots$ 。



$$A_i = \frac{1}{\pi} \int_0^{2\pi} y(t) \cos i\omega t \cdot d\omega t = \frac{1}{\pi} \left[\int_{-\pi}^0 y(t) \cos i\omega t \cdot d\omega t + \int_0^{\pi} y(t) \cos i\omega t \cdot d\omega t \right]$$

右端第一项 = $\frac{1}{\pi} \int_{\pi}^0 y(-t) \cdot \cos(-i\omega t) \cdot d(-\omega t) = \frac{1}{\pi} \int_{\pi}^0 y(t) \cos i\omega t \cdot d(\omega t)$

$$= -\frac{1}{\pi} \int_0^{\pi} y(t) \cos i\omega t \cdot d\omega t$$

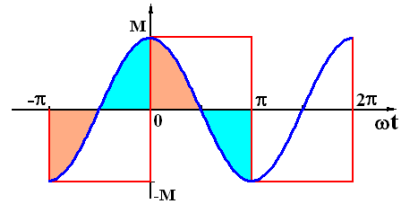
$$= \frac{1}{\pi} \left[-\int_0^{\pi} y(t) \cos i\omega t \cdot d\omega t + \int_0^{\pi} y(t) \cos i\omega t \cdot d\omega t \right] = 0$$

$$B_1 = \frac{1}{\pi} \int_0^{2\pi} y(t) \cdot \sin \omega t \cdot d\omega t = \frac{4}{\pi} \int_0^{\frac{\pi}{2}} M \sin \omega t \cdot d\omega t = \frac{4M}{\pi} [-\cos \omega t]_0^{\frac{\pi}{2}} = \frac{4M}{\pi}$$

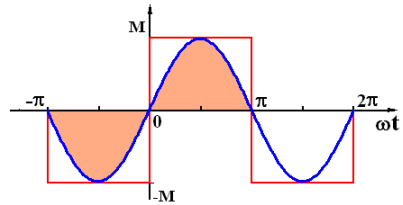
$$y_1(t) = A_1 \cos \omega t + B_1 \sin \omega t = \frac{4M}{\pi} \sin \omega t$$

$$\begin{cases} Y_1 = \sqrt{A_1^2 + B_1^2} = B_1 = \frac{4M}{\pi} \\ \varphi_1 = \arctan \frac{A_1}{B_1} = 0 \end{cases}$$

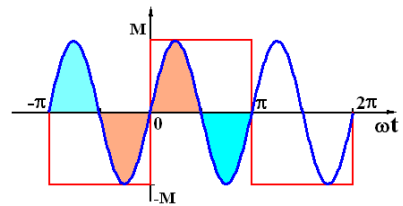
$$A_i = \frac{1}{\pi} \int_0^{2\pi} y(t) \cdot \cos i\omega t \cdot d\omega t = 0$$



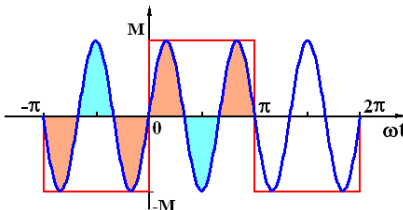
$$B_1 = \frac{1}{\pi} \int_0^{2\pi} y(t) \cdot \sin \omega t \cdot d\omega t = \frac{4M}{\pi}$$



$$B_2 = \frac{1}{\pi} \int_0^{2\pi} y(t) \cdot \sin 2\omega t \cdot d\omega t = 0$$

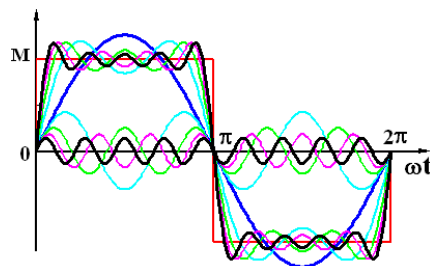
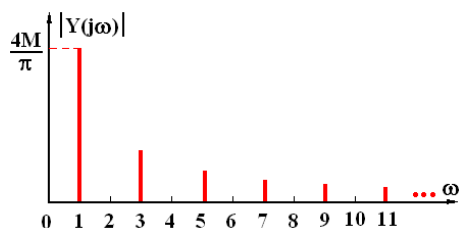


$$B_3 = \frac{1}{\pi} \int_0^{2\pi} y(t) \cdot \sin 3\omega t \cdot d\omega t = \frac{1}{3} \cdot \frac{4M}{\pi}$$



$$y(t) = A_0 + \sum_{n=1}^{\infty} Y_n \sin n\omega t = y_0(t) + y_1(t) + y_2(t) + y_3(t) + \dots$$

$$= \frac{4M}{\pi} \left[\sin \omega t + \frac{1}{3} \sin 3\omega t + \frac{1}{5} \sin 5\omega t + \frac{1}{7} \sin 7\omega t + \dots \right]$$



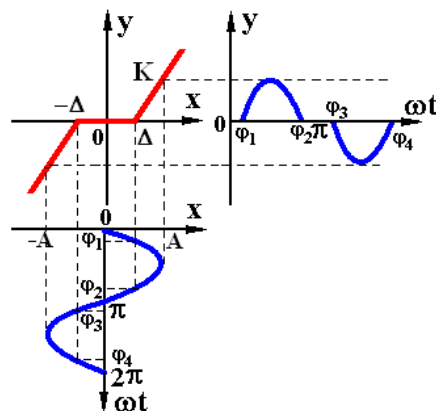
理想继电特性的描述函数

$$N(A) = \frac{B_1}{A} = \frac{4M}{\pi A}$$

2. 死区特性的描述函数

根据死区特性的奇对称性有：

$$\begin{cases} A_0 = 0 \\ A_1 = 0 \end{cases}$$



$$\begin{aligned} B_1 &= \frac{1}{\pi} \int_0^{2\pi} y(t) \cdot \sin \omega t \cdot d\omega t = \frac{2}{\pi} \int_0^{\pi} y(t) \cdot \sin \omega t \cdot d\omega t \\ &= \frac{2}{\pi} K \int_{\varphi_1}^{\varphi_2} (A \sin \omega t - \Delta) \cdot \sin \omega t \cdot d\omega t \\ &= \frac{4}{\pi} \int_{\varphi_1}^{\pi/2} K(A \sin \omega t - \Delta) \cdot \sin \omega t \cdot d\omega t \\ &= \frac{4KA}{\pi} \int_{\varphi_1}^{\pi/2} \sin^2 \omega t \cdot d\omega t - \frac{4K\Delta}{\pi} \int_{\varphi_1}^{\pi/2} \sin \omega t \cdot d\omega t \\ &= \frac{4KA}{\pi} \int_{\varphi_1}^{\pi/2} \frac{1}{2} (1 - \cos 2\omega t) \cdot d\omega t - \frac{4K\Delta}{\pi} \int_{\varphi_1}^{\pi/2} \sin \omega t \cdot d\omega t \\ &= \frac{4KA}{\pi} \left[\frac{\omega t}{2} - \frac{1}{4} \sin 2\omega t + \frac{\Delta}{A} \cos \omega t \right]_{\varphi_1}^{\pi/2} \end{aligned}$$

$$A \sin \varphi_1 = \Delta \quad \Rightarrow \quad \sin \varphi_1 = \frac{\Delta}{A} \quad \Rightarrow \quad \varphi_1 = \arcsin \frac{\Delta}{A}$$

$$\begin{aligned} B_1 &= \frac{4KA}{\pi} \left[\frac{\pi}{4} - \frac{1}{2} \arcsin \frac{\Delta}{A} + \frac{1}{2} \frac{\Delta}{A} \cos(\arcsin \frac{\Delta}{A}) - \frac{\Delta}{A} \cos(\arcsin \frac{\Delta}{A}) \right] \\ &= \frac{4KA}{\pi} \left[\frac{\pi}{4} - \frac{1}{2} \arcsin \frac{\Delta}{A} - \frac{\Delta}{2A} \sqrt{1 - \left(\frac{\Delta}{A}\right)^2} \right] \quad A \geq \Delta \end{aligned}$$

死区特性的描述函数

$$N(A) = \frac{B_1}{A} = \frac{2K}{\pi} \left[\frac{\pi}{2} - \arcsin \frac{\Delta}{A} - \frac{\Delta}{A} \sqrt{1 - \left(\frac{\Delta}{A}\right)^2} \right] \quad A \geq \Delta$$

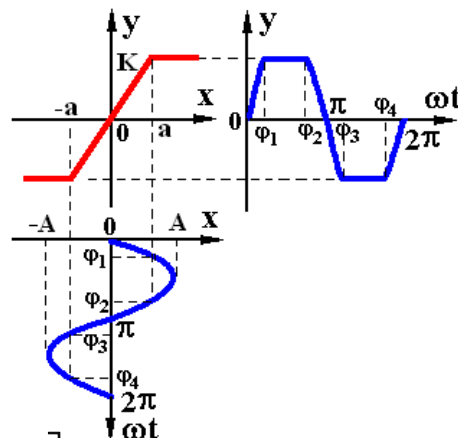
3. 饱和特性的描述函数

$$y(t) = \begin{cases} KA \sin \omega t & 0 \leq \omega t \leq \varphi_1 \\ Ka & \varphi_1 \leq \omega t \leq \pi/2 \end{cases}$$

利用饱和特性的奇对称性

$$A \sin \varphi_1 = a, \quad \varphi_1 = \arcsin \frac{a}{A}$$

$$\begin{aligned} B_1 &= \frac{4}{\pi} \left[\int_0^{\varphi_1} KA \sin^2 \omega t \cdot d\omega t + \int_{\varphi_1}^{\pi/2} Ka \sin \omega t \cdot d\omega t \right] \\ &= \frac{4KA}{\pi} \left[\int_0^{\varphi_1} \frac{1}{2} (1 - \cos 2\omega t) \cdot d\omega t + \frac{a}{A} \int_{\varphi_1}^{\pi/2} \sin \omega t \cdot d\omega t \right] \\ &= \frac{4KA}{\pi} \left\{ \left[\frac{1}{2} \omega t - \frac{1}{4} \sin 2\omega t \right]_0^{\varphi_1} + \left[\frac{a}{A} (-\cos \omega t) \right]_{\varphi_1}^{\pi/2} \right\} \\ &= \frac{4KA}{\pi} \left[\frac{1}{2} \arcsin \frac{a}{A} + \frac{a}{2A} \sqrt{1 - \left(\frac{a}{A}\right)^2} \right] \quad A \geq a \end{aligned}$$



饱和特性的描述函数

$$N(A) = \frac{B_1}{A} = \frac{2K}{\pi} \left[\arcsin \frac{a}{A} + \frac{a}{A} \sqrt{1 - \left(\frac{a}{A}\right)^2} \right] \quad A \geq a$$

4. 带滞环继电特性的描述函数

由右图有

$$\varphi_1: A \sin \varphi_1 = h$$

$$\sin \varphi_1 = \frac{h}{A},$$

$$\cos \varphi_1 = \sqrt{1 - \left(\frac{h}{A}\right)^2}$$

$$\varphi_2: A \sin \varphi_2 = mh$$

$$\sin \varphi_2 = \frac{mh}{A},$$

$$\cos \varphi_2 = -\sqrt{1 - \left(\frac{mh}{A}\right)^2}$$

$$\varphi_3: A \sin \varphi_3 = -h$$

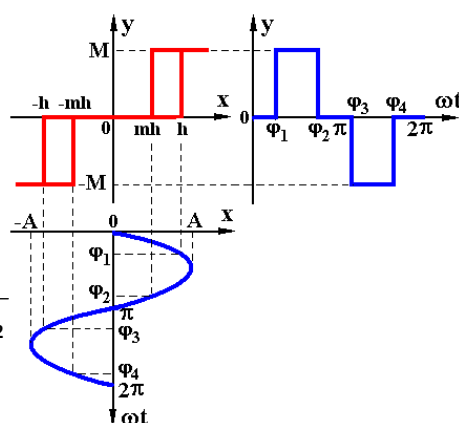
$$\sin \varphi_3 = \frac{-h}{A},$$

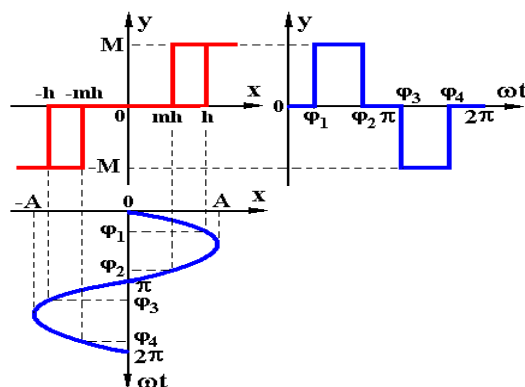
$$\cos \varphi_3 = -\sqrt{1 - \left(\frac{h}{A}\right)^2}$$

$$\varphi_4: A \sin \varphi_4 = -mh$$

$$\sin \varphi_4 = \frac{-mh}{A},$$

$$\cos \varphi_4 = \sqrt{1 - \left(\frac{mh}{A}\right)^2}$$





$$A_1 = \frac{1}{\pi} \int_0^{2\pi} y(t) \cdot \cos \omega t \cdot d\omega t = \frac{1}{\pi} \left[\int_{\varphi_1}^{\varphi_2} M \cos \omega t \cdot d\omega t - \int_{\varphi_3}^{\varphi_4} M \cos \omega t \cdot d\omega t \right]$$

$$= \frac{M}{\pi} \left\{ [\sin \omega t]_{\varphi_1}^{\varphi_2} - [\sin \omega t]_{\varphi_3}^{\varphi_4} \right\} = \frac{M}{\pi} \left[\frac{h}{A} (m-1) + \frac{h}{A} (m-1) \right] = \frac{2Mh}{\pi A} (m-1)$$

$$B_1 = \frac{1}{\pi} \int_0^{2\pi} y(t) \cdot \sin \omega t \cdot d\omega t = \frac{1}{\pi} \left[\int_{\varphi_1}^{\varphi_2} M \sin \omega t \cdot d\omega t - \int_{\varphi_3}^{\varphi_4} M \sin \omega t \cdot d\omega t \right]$$

$$= \frac{M}{\pi} \left\{ [-\cos \omega t]_{\varphi_1}^{\varphi_2} + [\cos \omega t]_{\varphi_3}^{\varphi_4} \right\}$$

$$= \frac{M}{\pi} \left\{ \sqrt{1 - \left(\frac{h}{A}\right)^2} + \sqrt{1 - \left(\frac{mh}{A}\right)^2} + \sqrt{1 - \left(\frac{h}{A}\right)^2} + \sqrt{1 - \left(\frac{mh}{A}\right)^2} \right\}$$

$$= \frac{2M}{\pi} \left[\sqrt{1 - \left(\frac{h}{A}\right)^2} + \sqrt{1 - \left(\frac{mh}{A}\right)^2} \right]$$

带滞环继电器特性的描述函数

$$N(A) = \frac{B_1}{A} + j \frac{A_1}{A} = \frac{2M}{\pi A} \left[\sqrt{1 - \left(\frac{h}{A}\right)^2} + \sqrt{1 - \left(\frac{mh}{A}\right)^2} \right] + j \frac{2Mh}{\pi A^2} (m-1)$$